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NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2025
Odd Semester - 2025-26
M101 - MATHEMATICS - I

TIME ALLOTTED: 3Hours **Maximum Marks : 70**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Part A

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. Cayley Hamilton theorem is **(1)** CO1 BL1 PO1
   - (a) Every square matrix satisfies its own characteristic equation
   - (b) Every symmetric matrix satisfies its own characteristic equation
   - (c) Every skew-symmetric matrix satisfies its own characteristic equation
   - (d) Every orthogonal matrix satisfies its own characteristic equation

2. The determinant of a triangular matrix is equal to **(1)** CO1 BL1 PO1
   - (a) Zero.
   - (b) Sum of diagonal element.
   - (c) Product of diagonal element.
   - (d) Trace of the matrix.

3. For a homogeneous function f(tx,ty)=t^nf(x,y) If n=0, f is: **(1)** CO1 BL1 PO1
   - (a) Linear
   - (b) Homogeneous of degree 1
   - (c) Homogeneous of degree 0 (scale invariant)
   - (d) Zero function only

4. The series $\sum_{n=1}^{\infty}\frac{1}{n^p}$ is convergent if **(1)** CO1 BL1 PO1
   - (a) $P \geq 1.$
   - (b) $P \leqslant 1.$
   - (c) P>1
   - (d) P<1

5. The maximal directional derivative of xy+yz+zx at (1,2,3) is **(1)** CO2 BL2 PO2

(a) $\sqrt{61}$. (b) $\sqrt{91}$.
(c) $5\sqrt{2}$. (d) $2\sqrt{5}$.

6. The sequence $1+\frac{1}{n}$ **(1)** CO2 BL2 PO2

(a) diverges (b) oscillates
(c) converges to the limit 1 (d) None of these

7. For small h, the linear approximation of f(x+h) is: **(1)** CO2 BL1 PO2

(a) f(x) (b) f(x)+f'(x)h

(c) f(x)+h (d) f'(x)

8. The series $\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+\frac{1}{4.5}+\ldots\ldots\ldots\ldots\ldots$ is convergent to the sum **(1)** CO2 BL2 PO2

(a) 0 (b) 1
(c) 2 (d) 3

9. The value of a for which $\vec{F}=\left(axy-z^3\right)\hat{i}+\left(a-2\right)x^2\hat{j}+\left(1-a\right)xz^2\hat{k}$ is irrotational is **(1)** CO2 BL2 PO2

(a) 2 (b) 0
(c) 4 (d) -4

10. If the matrix $\begin{pmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ \lambda & -3 & 0 \end{pmatrix}$ is singular, then the value of λ is **(1)** CO2 BL2 PO1

(a) 0. (b) 4.
(c) 2. (d) -1.

11. Find the value of the constant m so that the vector $\left(mxy-z^3\right)\hat{i}+\left(m-2\right)x^2\hat{j}+\left(1-m\right)xz^2\hat{k}$ is irrotational. **(1)** CO1 BL1 PO1

(a) 0. (b) 4.
(c) -4. (d) None

12. What is the value of the double integral $\int_0^1\int_0^{y^2} xydxdy$? **(1)** CO2 BL2 PO1

(a) $\frac{1}{15}$. (b) $\frac{1}{12}$.
(c) $\frac{1}{17}$. (d) None

Part B
(Answer any three of the following) 3x5=15

13. Check whether Rolle's theorem is applicable or not for the function f(x)= mod x defined on [−1,1]. Justify your answer.	**(5)**	CO3	BL1	PO3
14. If $f(x,y)=\begin{cases}\dfrac{xy(x^2-y^2)}{(x^2+y^2)}; & (x,y)\neq(0,0)\\ 0 & ;(x,y)\neq(0,0)\end{cases}$ find the partial derivatives $f_x \quad and \quad f_y$ at the point (0,0).	**(5)**	CO3	BL1	PO3
15. Determine Rank for the matrix $A=\begin{pmatrix}1 & -1 & 2 & 0 & 4\\ 2 & 3 & 1 & 5 & 2\\ 1 & 3 & -1 & 0 & 3\\ 1 & 7 & -4 & 1 & 1\end{pmatrix}$	**(5)**	CO3	BL3	PO3
16. Test the series for convergence $\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\frac{4}{2^4}+\ldots\ldots\ldots\ldots$	**(5)**	CO3	BL2	PO3
17. Using divergence theorem , evaluate $\int_S\int \vec{F}.\hat{n} \quad dS$, where $\vec{F}=4xz\hat{i}+xyz^2\hat{j} \quad +3z\hat{k}$ above the xy-plane bounded by the cone $z^2=x^2+y^2$ and the plane z=4	**(5)**	CO4	BL3	PO4

Part C

(Answer any three of the following) 3x15=45

18. (a) Use Mean value theorem prove that $0<\dfrac{1}{\log(1+x)}-\dfrac{1}{x}<1$, for x> 0.	**(5)**	CO2	BL2	PO4
(b) Show that $1-\dfrac{1}{2^2}+\dfrac{1}{3^2}-\dfrac{1}{4^2}+\ldots\ldots$ is convergent series.	**(5)**	CO3	BL3	PO4
(c) Verify Cauchy's Mean Value Theorem for the following pair of functions $f(X)=\sqrt{x}$, $g(x)=1/\sqrt{x}$; $x\in[1,2]$	**(5)**	CO4	BL4	PO10

19.

(a) Given the system of equations : **(10)** CO4 BL5 PO12

$$x_1 + 4x_2 + 2x_3 = 1$$
$$2x_1 + 7x_2 + 5x_3 = 2k$$
$$4x_1 + mx_2 + 10x_3 = 2k + 1,$$

find for what values of k and m the system has (i) a unique solution ,(ii) no solution, (iii) many solutions.

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(b) Find the inverse of the matrix $\begin{pmatrix} 1 & -2 & 1 \\ 2 & 1 & -1 \\ 1 & -3 & 2 \end{pmatrix}$. **(5)** CO3 BL6 PO10

20. (a) Verify that the matrix $\begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix}$ is diagonalisable. If so, find the diagonal matrix which will diagonalise it. **(7)** CO2 BL5 PO5

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(b) Reduce the matrix $A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 0 & 0 & 2 & 2 \\ 3 & 9 & 6 & 10 \\ 2 & 6 & 2 & 6 \end{pmatrix}$ to an equivalent echelon matrix and hence find its rank. **(8)** CO2 BL6 PO4

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21. (a) Evaluate $\oint \{ (x^2 - \cosh y)\, dx + (y + \sin x)\, dy \}$ by Green's theorem, where C is the rectangular region with vertices $(0,0), (\pi,0), (\pi,1), (1,0)$. **(7)** CO3 BL3 PO1

(b) Use divergence theorem to evaluate $\int\int_S \vec{F}.\hat{n}\, dS$, where $\vec{F} = 3xz\hat{i} + y^2\hat{j} - 3yz\hat{k}$ and S is the surface of the cube bounded by the planes $x = 0, y = 0, z = 0$ and $x = 1, y = 1, z = 1$. **(8)** CO4 BL4 PO2

22. (a) Evaluate $\int\int\int (p + q + r + 1)^4 dp dq dr$ over 3D region $p, q, r \geq 0, p + q + r \leq 1$ **(10)** CO4 BL2 PO4

(b) Change the order of integration and hence evaluate $\int_0^1 dy \int_y^1 e^{x^2}\, dx$. **(5)** CO3 BL2 PO3